Subject Description Form

Subject Code	ME5304
Subject Title	Soft Robotics
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject aims to provide students with comprehensive knowledge of advanced bio-inspired methods in soft robotics and deformable machines, including its actuators, sensors, structures, manufacturing, modelling, simulation and automatic control.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Understand the advantages and operation principles behind soft robotic systems; b) Identify scenarios where soft robotic structures are better suited than conventional robots; c) Design, manufacture, and automatic control of soft robots; d) Apply mathematical modelling and simulation techniques for soft robots.
Subject Synopsis/ Indicative Syllabus	**Functional and Intelligent Soft Robots:** Differences between rigid and soft robot mechanisms; Types of soft materials used in soft robotics; Properties and fabrication; Applications of soft robotic systems. **Soft Actuators and Soft Sensors:** Types of soft actuators and their principles; Design considerations and fabrication of soft actuators; Types of soft sensor technologies; Materials and sensing principles for soft robot sensors; Cases of study. **Morphology Design and Manufacture:** Design principles; Bio-inspired soft designs; Variable morphology robots; Continuum manipulators; Origami inspired structures; Inflatable and complaint systems; Tools for soft robotic design; Fabrication techniques of soft components; Hybrid soft-rigid robots. **Modelling, Analysis and Control:** Mathematical modelling principles of soft structures; Types of dynamic models; Analytical and computational shape representation methods; Simulation of deformable robotic systems; Stability of soft robots; Motion and shape control strategies for soft robots. **Applications and Future Directions:** Applications in healthcare, service, and manufacturing; Bio-inspired robotic systems; Societal and environmental implications; Challenges and future trends. **Practical Work:** Examples and demonstrations of soft robotic technologies; Group projects; Cases of study to integrate and reinforce the knowledge gained in this subject.

Teaching/Learning Methodology	1. Lectures aim at providing students with fundamental knowledge required for understanding and analysing different soft robotic systems, including their pros and cons, targeted applications, manufacture and control techniques, and modelling and simulation methods (Outcomes a to d). 2. Tutorials aim at enhancing students' analytical and problem-solving skills on soft robotics. Students will be able to solve real-world problems using the knowledge they acquired in the class (Outcomes a to d). 3. The experiments/project aim to provide hands-on experience for developing soft robots, and reinforcing the acquired knowledge. (Outcomes a to d)

Teaching/Learning Methodology	**Intended Subject Learning Outcomes to be assessed**			
	a	**b**	**c**	**d**
1. Lecture	✓	✓	✓	✓
2. Assignments	✓	✓	✓	✓
3. Project	✓	✓	✓	✓

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	**% weighting**	**Intended subject learning outcomes to be assessed**			
		a	**b**	**c**	**d**
1. Assignments	20%	✓	✓	✓	✓
2. Project	30%	✓	✓	✓	✓
3. Test	10%	✓	✓	✓	✓
4. Final Exam	40%	✓	✓	✓	✓
Total	100 %				

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Overall Assessment:

0.40 x End of subject exam + 0.60 x Continuous assessment.

1. The continuous assessment aims at evaluating the progress of the students' study, assisting them in self-monitoring the respective learning outcomes, and applying the knowledge learnt in practical situations.
2. The examination is used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes

Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorials / Exercises / Laboratory	6 Hrs.
	Other student study effort:	
	▪ Assignments and project	40 Hrs.
	▪ Self-study	31 Hrs.
	Total student study effort	110 Hrs.
Reading List and References	Textbooks: 1. The Science of Soft Robots: Design, Materials and Information Processing, Springer Publishing, 2023 2. Handbook on Soft Robotics, Springer Publishing, 2024. 3. Soft Robotics: Transferring Theory to Applications, Springer Publishing, 2015 4. Soft Actuators: Materials, Modeling, Applications, and Future Perspectives, Springer Publishing, Second Edition, 2019.	
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